OPEN ELECTROTHERMAL BENCH / EBT-R01

When Surface Temperatures Mislead

Computational feasibility study for measurement observability and cross-boundary validation of compact electrothermal models

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STATUS: SYNTHETIC RESULTS ONLY. No physical measurements, equipment calibration, institutional approval, KTH sponsorship, or experimental validation has occurred. The numerical results demonstrate problem feasibility and software verification, not the thermal behaviour of any real MOSFET.

Purpose of this dossier

Establish a credible, testable research problem before contacting KTH: working code, a physical conservation model, independently held-out numerical scenarios, a targeted literature assessment, an instrumented bench concept and explicit research/approval gates.

Companion reproducibility package: source code, seven automated tests, synthetic CSV outputs, figures, metrics, study protocol and bibliography. Code MIT; original documentation and generated synthetic data CC BY 4.0.

Executive technical summary

The proposed paper studies a measurement-design problem: can package and heatsink sensors adequately constrain the internal temperatures predicted by compact electrothermal models, and can a model identified under one thermal boundary predict a different cooling or interface configuration? The present numerical benchmark is deliberately small and transparent. It is a method-shakedown, not evidence of novelty or hardware accuracy.

0.104 °C	maximum difference between two synthetic case-temperature traces
3.75 °C	maximum hidden junction-temperature difference for those same models
6.79 °C	forced-convection holdout RMSE: fixed-boundary 1RC model
0.57 °C	same holdout RMSE: 1RC with independently supplied synthetic boundary R

What these numbers establish

• A pair of **synthetically defined** assemblies can have similar externally observable temperatures but different internal states.

• An unchanged boundary model may fail on a different cooling configuration even when its original fit is respectable.

• Updating a boundary resistance known independently by construction can materially improve a synthetic holdout; this is **not** a verified real-world measurement strategy.

• The benchmark verifies equations and proposes where physical instrumentation will be needed. It does **not** establish junction-temperature accuracy for a physical device.

Primary research risk

Theoretical identifiability and thermal model validation are extensively covered in prior work. Scientific novelty remains unproven. The most promising differentiated output would be a properly controlled open experimental benchmark that quantifies cross-assembly transfer, measurement observability and uncertainty together.

1. Physical model and numerical verification

The synthetic model consists of three energy-storing nodes: internal junction j, sensor-accessible package p and heat sink s. The heat path includes junction-to-package Rjp, package-to-sink Rps and sink-to-ambient Rsa; input power P is applied at j. All node heat capacities and resistances are positive.

Governing balances

Cj dTj/dt = P - (Tj - Tp)/Rjp

Cp dTp/dt = (Tj - Tp)/Rjp - (Tp - Ts)/Rps

Cs dTs/dt = (Tp - Ts)/Rps - (Ts - Ta)/Rsa

Constants: Rjp = 0.75 K/W, Rps = 0.45 K/W, Rsa = 3.00 K/W; Cj = 0.60 J/K, Cp = 9.0 J/K, Cs = 70 J/K; Ta = 24 °C. A 5 W step runs for 750 s, then power is removed. A 5 s sampled exact zero-order-hold propagator eliminates dependence on a numerical ODE time step at sampling boundaries.

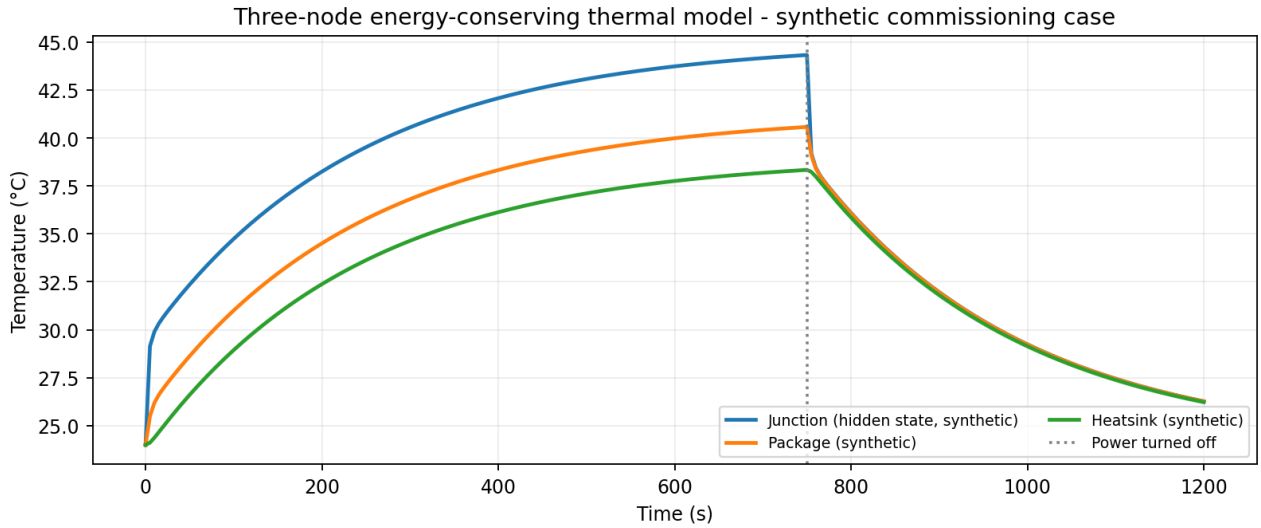


Figure 1. All lines are generated by the numerical model. The junction trace is a simulated hidden state, not a measurement.

Verification activities passed

Seven automated tests verify zero-power ambient equilibrium, analytical steady-state limit, passive temperature ordering, grid-invariant sampled solution, approximate energy conservation, the single-RC steady-state formula and cooling response to changed thermal boundary. The total relative discrete energy-balance residual is 1.64e-06 at a 1 s integration grid. This is a solver/implementation check only.

2. Identifiability counterexample

We compare two otherwise identical synthetic systems: A with Rjp = 0.75 K/W, and B with Rjp = 1.50 K/W. Both have the same input power, ambient temperature, case/sink resistances and capacities. The package is assumed to be the instrumented observable; the junction is inaccessible to a surface thermometer.

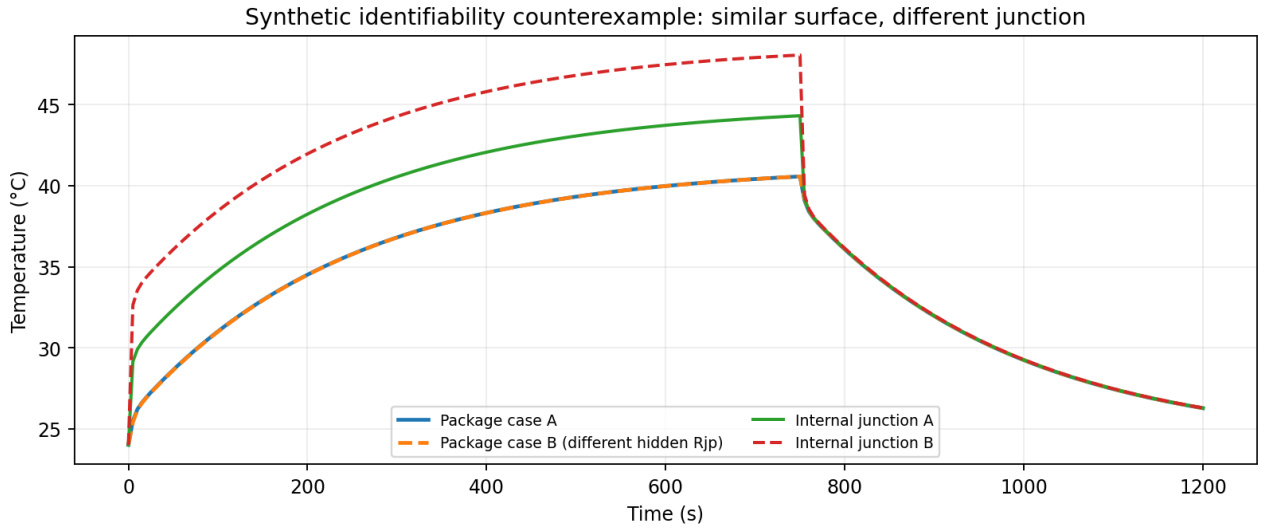


Figure 2. Both package-temperature trajectories closely overlap, whereas the modelled internal junction trajectories diverge.

Across the prescribed time history, the maximum package discrepancy is **0.104 °C** while the maximum junction difference is **3.75 °C**. The synthetic measurement-noise standard deviation is assumed to be 0.15 °C. Hence the maximum package change is smaller than a single assumed noise standard deviation, even though internal temperature differs by several degrees. This does not prove global parameter non-identifiability or define a detection probability: those require a formal likelihood, noise-correlation and power analysis.

Implication for instrument selection

A package thermocouple and thermal camera both measure surface quantities. Agreement between them cannot, by itself, validate junction temperature. Future experiment designs need either a qualified junction-sensitive electrical measurement, an externally constrained inner thermal path, or an explicitly bounded junction uncertainty. This limitation must appear in all claims.

3. Holdout prediction under changed cooling

A single first-order RC model is fitted only to a synthetic noisy package temperature measured during 5 W natural cooling. Independent synthetic cases are reserved for evaluation: a 3 W power change, forced convection (Rsa = 0.90 K/W) and a changed interface (Rps = 1.00 K/W). The 0.15 °C sensor-noise assumption is used for all nominal noisy traces.

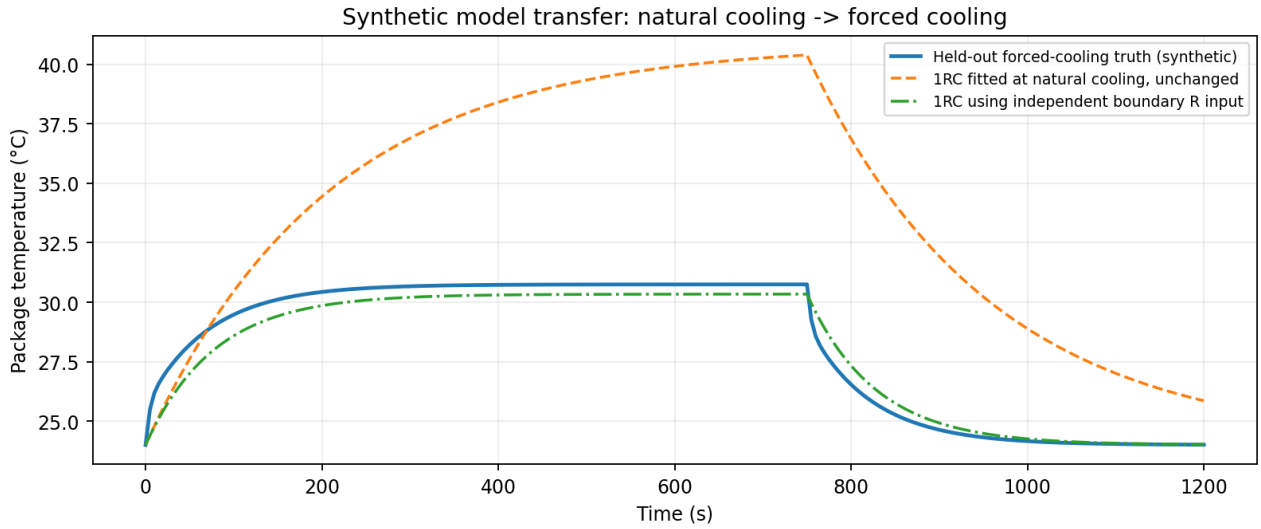


Figure 3. In this synthetic holdout, an unchanged lumped thermal boundary overpredicts temperatures under forced convection. Supplying an independently known boundary R improves predictions.

Test condition	Evaluation model	RMSE (°C)
5 W natural (calibration)	fixed 1RC	0.466
3 W natural (holdout)	fixed 1RC	0.313
5 W forced (holdout)	fixed 1RC	6.788
5 W forced (same holdout)	boundary-aware 1RC	0.568
5 W modified interface (holdout)	unadapted fixed 1RC	2.157

The boundary-aware model changes an independently supplied Rsa input, not a parameter inferred from held-out case temperatures. That is a necessary anti-leakage condition, but in hardware Rsa will itself require measurement and uncertainty. These results demonstrate a feasible research methodology, not comparative accuracy of actual physical models.

4. Cross-case model performance

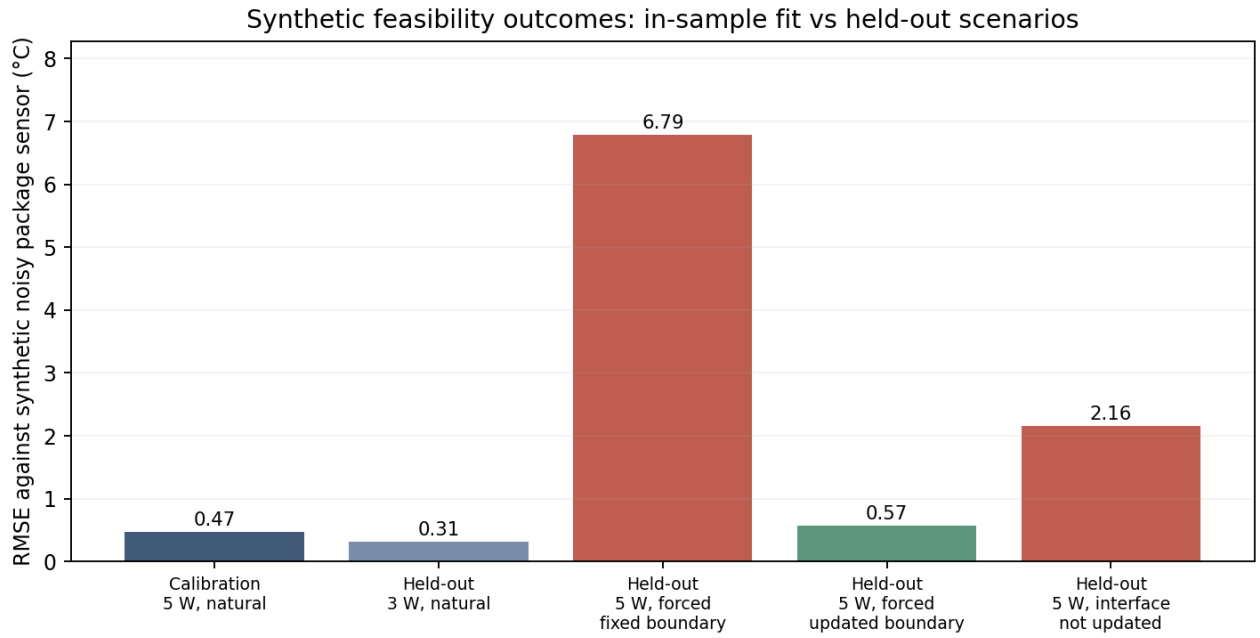


Figure 4. Errors compare model outputs with simulated noisy package data only; held-out scenarios remain excluded from calibration.

Interpretation and limitations

• Calibration RMSE is 0.466 °C, greater than the 0.15 °C injected noise scale: there is structural mismatch between a one-state compact model and a three-node system.

• The forced-convection error changes substantially with boundary information. This alone is not novel; it motivates independent boundary characterization and fair comparisons.

• Modified interface behavior remains inadequately modeled: an airflow-only correction does not resolve mounting/thermal-contact changes.

• The synthetic truth model has fixed linear parameters, deterministic power delivery, known ambient and independent Gaussian sensor noise. Real devices introduce nonlinearity, measurement lag, contact variation, airflow uncertainty and electrical loss dynamics.

Evidence required to upgrade to a paper

The physical campaign must predefine held-out hardware configurations; independently characterize boundary and mounting conditions; evaluate a qualified temperature measurement system; document variation across genuine remounts; and report predictive performance with uncertainty and limits. A specific narrow contribution must survive a full-paper literature review.

5. Draft experimental campaign (not authorized)

Use an appropriately rated low-voltage MOSFET assembly and a separate resistive reference heater. Provide a repeatable mechanical mount, appropriate heat sink, controlled source, measured dissipation, ambient reference, at least package and sink temperature sensors, and synchronized data acquisition. An internal junction-temperature reference is conditional on technical feasibility and suitable calibration. Do not equate case temperature with junction temperature.

Factor	Candidate structure	Reason
Input dissipation	3 safely qualified levels	Estimate load dependence and boundary transfer
Cooling	3 regimes, monitored	Separate natural and forced boundary conditions
Thermal interface	2 defined assembly configurations	Quantify contact/assembly sensitivity
Replicates	3 independent remounts per configuration	Estimate assembly repeatability
Initial envelope	54 candidate independent runs	Pilot must confirm statistical precision and time budget
Withheld validation	Entire cooling/mounting configurations	Prevent time-series leakage from correlated samples

Laboratory prerequisites

• Written facility permission for energized testing; do not perform energized tests in manufacturing-only workshops.

• Approved schematic, current/voltage/safe operating area review, current limit, protection and emergency shutdown.

• Sensor and instrument identification with calibration evidence/qualified verification, synchronized timestamps, uncertainty budget.

• A written operator procedure with temperature stop limits, supervised test scope and failed-run preservation.

• Permission for publishing allowed photographs, manufacturer information and non-confidential facility records.

Pilot go/no-go criteria

Proceed to a main campaign only when the chosen circuit is demonstrably safe under expected worst-case conditions, sensor lag and repeatability are quantified, test time is feasible, the intended boundary can be measured independently, and the research objective is not already fully resolved by accessible literature.

6. Relevant prior art and novelty controls

Source	Existing contribution	What we must NOT claim
Putthong et al. (2026), DOI 10.53941/perat.2026.100014	SVD/information-sufficient design and held-out predictions on electrothermal MOSFET models	Novelty of experiment design or out-of-sample assessment alone
Schulz et al. (2026), IEEE TPEL, accepted	Resolvable Cauer network layers, profile likelihood	Discovery of Cauer non-identifiability
Zhang et al. (2024), DOI 10.1109/TPEL.2024.3367854	Regularized transient thermal parameter identification	Novelty of sparse thermal identification
Ress et al. (2026), DOI 10.3390/en19112678	Improved thermal transient measurements for wide-bandgap devices	New thermal transient measurement concept absent comparison
Bhadoria (2025), KTH PhD thesis	SiC cooling, microchannels, thermal and overcurrent response	Novel SiC cooling comparison
Cnudde et al. (2026), THERMINIC	Experimental lumped spreading-resistance model validation	Novelty of lumped electronics thermal validation alone

Current defensible positioning: an open protocol and benchmark studying the joint measurement/transfer/uncertainty problem in a specific accessible cooling assembly. This is a **potential contribution**, not a verified gap. Full-text review, systematic citation search and independent scholarly feedback are required before using words such as "novel" in an abstract.

Technical framework references

ISO/IEC 17025:2017 provides laboratory competence principles (no accreditation claimed). JCGM 100:2008 and its 2026 amendment provide an uncertainty framework. ASME V&V; 20-2009 (R2021) offers concepts for specified validation points in CFD and heat transfer. JEDEC JESD51 methods may be applicable to device-level thermal measurement, but no JEDEC procedure has been performed here.

7. Reproduction, open data and publication path

Deliverable package already available

The accompanying archive contains a physics solver, exact sampled propagator, 7 executable tests, reproducible simulation script, generated CSV results, integrity hashes, four plots, scope/governance documents, draft physical test procedure and an explicit literature-gap tracker. All provided data are synthetic and are labeled as such at file and column level.

Reproduction commands: create a Python environment; install requirements; install the project editable; run `python -m pytest -q` followed by `python scripts/run_study.py`. The random seed is fixed at 20261105. Verify expected JSON results in `results/metrics.json`. The synthetic CSV and JSON outputs are hashed in `results/SHA256SUMS.json`.

Next phases

Gate	Evidence needed
G1: Scientific gap	Full-text literature matrix and bounded contribution reviewed by a researcher
G2: Physical feasibility	Authorized lab, supported device, qualified sensors and power system
G3: Pilot	Observed repeatability, verified safety envelope and realistic runtime
G4: Primary experiment	Replicated measured data, blind holdouts and uncertainty analysis
G5: Paper readiness	External methodological critique, clean reproduction, scientific interpretation
G6: Public release	Permissions, GitHub, Zenodo DOI, honest technical post; scholarly venue if merited

KTH contact after tangible progress

The current package is enough to support a serious **methodology discussion**, but not yet a publication proposal based on empirical findings. The most valuable next evidence is an independently characterized benchtop data set that demonstrates where surface-only observations succeed or fail. Collaboration cannot be assumed: authorship, supervision, permissions and data rights must be agreed explicitly.

Reference links (selected)

Putthong et al. (2026): https://doi.org/10.53941/perat.2026.100014

Schulz et al. (2026): https://erickschulz.dev/publications/src/data/publications/identifiability-cauer-thermal-networks/

Zhang et al. (2024): https://doi.org/10.1109/TPEL.2024.3367854

Ress et al. (2026): https://doi.org/10.3390/en19112678

JCGM guides: https://www.bipm.org/en/committees/jc/jcgm/publications

ASME V&V; 20: https://www.asme.org/codes-standards/find-codes-standards/standard-for-verification-and-validation-in-computational-fluid-dynamics-and-heat-transfer

ISO 17025: https://www.iso.org/standard/66912.html